

Do Male Rats Share in the Parental Care of Pups?

Tessa Peck: 300455023

Supervisor: Bart Ellenbroek

Victoria University of Wellington

489 Research Project

Word Count: 7977

Acknowledgements: Thank you to Bart Ellenbroek and Jiun Youn for mentorship. Jenny Berrio-Sanchez, Michaela Pettie and Meyrick Kidwell for support and guidance.

Abstract

The current study aimed to investigate the influence of paternal care on offspring's emotional behaviour. Infant and parental attachment was investigated using infant ultrasonic vocalisation (USV) potentiation and father interest experiments. Offspring emotional behaviour was measured using novelty suppressed feeding (NSF), a modified successive alleys (mSAT), and anticipatory pleasure paradigms. Significant findings from infant USV, NSF and anticipatory pleasure paradigms suggest that presence of sires during development has a positive impact on offspring's emotional behaviour, at least in infancy and adolescence. Novel findings in the mSAT produced significant sex differences as seen in other models of rodent conflict anxiety. Together, these findings suggest that presence of sires during pregnancy and rearing has a substantial effect on offspring development and behaviour. Therefore, the breeding scheme used in experiments where maternal care may be compromised warrants consideration.

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

Early life experiences can profoundly affect adult behaviour. Children exposed to stressful environments are at higher risk for anxiety in adulthood (Heim & Nemeroff, 2001). A contributor to stress during childhood is quality of parental care. Poor parental bonding can increase risk of anxiety later in life (Darlene, Champagne, Liu, & Meaney, 1999). In children, parental care influences cognitive development and stress reactivity in adulthood (Liu, Diorio, Day, Francis, & Meaney, 2000; Francis & Meaney, 1999).

In monogamous rodents, such as rats and mice, parental care is solely provided by the mother. Like humans, maternal care in rats affects cognitive development in offspring (Whishaw & Kolb, 2014) and influences adult emotional behaviour (Caldji, Tannenbaum, Sharma, Francis, Plotsky, & Meaney, 1998). Maternal care provides a link between pups and their environment (Darlene, et al., 1999). Therefore, subtle modifications in maternal care can have large-scale effects on pup development, by changing the makeup of their whole environment and their brains. Experimental manipulations, such as drug-induced and genetic modifications, can reduce maternal care (Bosch & Neumann, 2008; Pederson, Vladlamundi, Bolliam, & Amico, 2006; Brumer & Buhot, 1999; Zhao & Li, 2009). This reduced care could confound behavioural data gathered from the offspring. Variability in maternal care could be ameliorated if sires contribute to care of offspring. However, the influence of paternal care on pups is not well understood as many animal facilities use intermittent breeding schemes that do not allow any contact between sires and offspring.

Parental Care

Maternal care is essential for infant survival (Kuroda, Tachikawa, Yoshida, Tsuneoka, Numan, 2011). Rats are born altricial so require a lot of care from birth (Broad, Curley, & Keverne (2006). Pups are reliant on their mother for protection, thermoregulation and nutrition (Kuroda et al., 2011). As such, they form strong bonds with their mother even if she

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

is abusive (Perry & Sullivan, 2014). Dams provide care through licking and grooming (LG), retrieval and arched back huddling (feeding) behaviours (Liu et al., 1997).

Maternal care in rats is stimulated by endocrine signals after pregnancy and birth (Champagne, Francis, Mar, & Meaney, 2003) and by ultrasonic vocalisations (USV) and odour cues from pups (Hofer, Masmela, Brunelli, & Shair, 1997). Generally, dams show a stable level of maternal care between litters (Champagne et al., 2003). However, maternal care can vary greatly between strains (Ehret, 2005) and individual dams (Champagne et al., 2003). These differences can be due to the quality of the maternal environment and the dams current emotional state (Ehret & Haack, 1984).

Parental care influences offspring's behavioural and endocrine response to stress in adulthood (Francis, Diorio, Liu, & Meaney, 1999). Higher maternal care leads to lower anxiety-like behaviour in the elevated plus maze (EPM; Priebe et al., 2005) and open field tests (Caldji et al., 1998). The mechanism through which this occurs is by LG (Francis et al., 1999), which can be performed by sires (Dewsbury, 1985). LG "programmes" behavioural responses in offspring (Caldji et al., 1998) by decreasing hypothalamic-pituitary-adrenal (HPA) responsivity to stress (Liu et al., 1997). Offspring of high LG mothers have decreased plasma adrenocorticotrophic hormone and corticosterone response to acute stress and increased hippocampal glucocorticoid receptor mRNA expression (Liu et al., 1997). Stress reactivity in adulthood is reduced through changes in expression of genes that regulate endocrine responses to stress (Francis et al., 1999) making the effects of maternal care on anxiety-like behaviour long lasting (Caldji et al., 1998).

Unlike dams, males are not spontaneously parental towards their offspring (Liu et al., 1997). While virgin females do not tend to be infanticidal, all male rats can show aggression towards offspring (Reisert et al., 2002) and avoid unfamiliar pups (Kuroda et al., 2011). However, males can perform all maternal behaviours except for lactation (Dewsbury, 1985).

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

Paternal behaviours rely on cohabitation with pups and presence of the dam (Kuroda et al., 2011; Ehret, 2005). Olfactory and auditory signals from dams can induce paternal behaviour in sires (Fujimoto, Liu, Lopatina, Brown, & Higashida, 2013) and paternal care in males is significantly reduced when these signals are blocked (Liu et al., 1997).

Animal Facility Breeding Schemes

Most animal facilities breed rats using a so-called intermittent breeding scheme where males are removed from pregnant females before birth, limiting the parental care to the mothers only. Alternatively, continuous breeding schemes are sometimes employed and involve the males and females cohabiting throughout pregnancy, delivery and rearing. Continuous housing with both parents could be beneficial for the offspring if sires contribute to the parental care of pups. In a comparison between breeding schemes Wright & Brown (2000) found that sires did not disrupt maternal care provided for the pups and performed LG behaviours just as often as the dam. Therefore, continuously bred offspring received twice as much stimulation as intermittently bred offspring. Offspring from continuous breeding schemes are also heavier at weaning, which is associated with increased wellbeing in adulthood (Allen, Dwinell, Zappa, Temple, & Thulin, 2013).

However, continuous breeding schemes increase the risk of infanticide. Breeding demands little energy from males compared to females (Li, Keverne, Aparicio, Ishino, Barton, & Surani, 1999). It is in their best interest to spread their resources over many litters to different females, in order to increase the chance of their genes passing on to future generations (Bateson, 1994). According to this, sires housed with a single female for their entire breeding lives should increase their investment in litters that they are housed with (Wright & Brown, 2000). Mice will exhibit paternal behaviours when housed with a single female but not multiple (Priestel & Young, 1978). Male rats also show reduced rates of infanticide when housed with pregnant dams and litters until weaning (Brown, 1986).

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

Continuous breeding schemes may affect pup development by influencing maternal care. Presence of the sire could reduce maternal stress and allow her to spend less time on some maternal behaviours, without abandoning the pups (Wright & Brown, 2000). However, continuous breeding could increase stress on the dam due to over-crowding (Allen et al., 2013). Increased housing density in adulthood is associated with increased anxiety-like behaviour in open field tests (Barker et al., 2017). Elevated stress hormones in the dam during the postpartum period will reduce maternal care provided for the pups (Brummette & Galea, 2010), transferring stress from mother to offspring.

Lastly, continuous breeding schemes are more draining on the dams resources. Dams enter postpartum estrus within 24 hours of giving birth (Conner & Davis 1980). The gestation and weaning periods are both approximately 3 weeks. Therefore, dams can be almost constantly pregnant if continuously housed with a male. This can be beneficial for animal facilities that demand a high throughput of animals. However, it can be costly for the dam, and potentially affect the quality of care she provides for her pups. Lactation and pregnancy demand a lot of energy (Li et al., 1999). The dam's energy reserves determines the quality of milk provided to her pups. In fact, pup growth rates are lower if dams have had litters in quick succession (Rodel, Prager, Stefanski, Holst, & Hudson, 2008). Unlike sires, who generally invest little energy into the upbringing of pups, dams must consider their current nutritional state and probability of having another litter when allocating resources to young (Shair, Muller, & Moore, 2008a). Therefore, dams cannot invest heavily in a current litter at the cost of their own resources which could be used for future litters.

Purpose of the Current Study

Studies by Meaney and colleagues have established the importance of parental care in offspring behaviour and development in both humans and rodents. In rats, the influence of parental care and early environment is primarily studied using dams only. Little work to date

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

has investigated the influence of paternal care. Previous work suggests that sires are capable of providing important and useful care for pups. The current study investigated the effect of paternal care on measures of infant and parental attachment, and emotional behaviour during infancy, adolescence and adulthood, by comparing continuous and intermittent breeding schemes.

It was hypothesised that sires housed continuously with their pups would contribute to parental care. Secondly, it was hypothesised that pups in continuous breeding schemes would exhibit higher levels of attachment to their fathers than those in intermittent breeding schemes. Thirdly, it was hypothesized that differences in parental care and attachment would induce long-term consequences in emotional behaviour in adulthood.

Infant and Parental Attachment

A parental-separation paradigm measured infant attachment to parents as well as indirectly measured sires contribution to parental care. If pups are placed alone in a novel environment they will emit 40 kHz (roughly between 30 and 40 kHz) calls (Branchi, Santucci, & Alleva, 2001). These arouse both male and female adult rats to retrieve pups (Ehret, 2005) and elicits maternal behaviour (Branchi et al., 2001). The presence of the dam and littermates provide thermal, textural, and odour stimuli that reduce calling rate (Hofer, 1996). All of these stimuli could potentially be provided by sires as well, although so far this has not been studied in rats. This would be particularly relevant in a continuous breeding scheme, where sires are constantly present and thus pups would be very familiar with the scent of the males.

USVs from pups provides an index of attachment as pups exhibit a 2-fold increase in USV rate when isolated after a brief reunification with dams or maternal cues (so-called maternal potentiation) but not litter mates (Shair, Muller, & Moore, 2008b; Hofer, 1996). The level of care provided to pups in their early development can alter patterns in USV.

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

Offspring of low LG dams emit more USVs when separated from mother and litter (Wohr & Schwarting, 2008). Assuming sires provide care for pups (Dewsbury, 1985), it is possible that pups will exhibit paternal potentiation of USVs following a brief exposure to sires as well. Therefore, patterns of USV potentiation can provide an indirect measure of a sires contribution to parental care. Previous work has found that USV response following contact with an adult depends on the pups rearing experience (Brunelli, Masmela, Shair, & Hofer, 1998). Pups reared without sires only exhibit potentiation to dams, and USV rate is suppressed following exposure to an adult male. Whereas those reared with sires will also show USV potentiation following contact with their own father. It was expected that continuously bred pups would produce USV potentiation following contact with sires but intermittently bred pups would not.

A father interest paradigm was used to measure sires attachment to pups. Liu et al. (2013) found that sires will begin to show a preference to a chamber with pups (compared to no pups) over the course of 3-10 minutes. It was expected that the strength of a males attachment to his pups would positively correlate with the time spent investigating his pups. Therefore, it was expected that continuously bred sires would spend more time investigating pups than intermittently bred sires, indicating a higher level of attachment.

Adolescence and Adulthood emotional behaviour

To assess whether breeding schemes affects offspring's emotional behaviour, rats were subjected to a modified version of the successive alleys test (mSAT; Deacon, 2013), which allowed for a repeated assessment of anxiety-like behaviour.

Adolescent and adult rats can differ in their responses to test of anxiety-like behaviour (Lynn & Brown, 2010). The HPA axis is still undergoing maturation during adolescence, so the different effects of breeding scheme could be more or less pronounced during adolescence compared to adulthood. Studies looking at the EPM, a similar measure of

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

conflict-anxiety, have reported conflicting results as to whether adolescents or adult rats exhibit more anxiety (Doremus-Fitzwater, Varlinskaya, & Spear, 2009; Doremus, Brunell, Varlinskaya, & Spear, 2003). Adolescent anxiety was measured using a novelty suppressed feeding (NSF) paradigm. This has similar underlying mechanisms as the SAT as the animal experiences the conflict of anxiety in a novel environment against the motivation to approach and eat an appetitive food.

Adult emotional behaviour was tested using an anticipatory pleasure paradigm. USVs from adults can measure positive and negative emotional states. 22kHz calls are emitted by rats during aversive situations and co-occur with freezing behaviours (Schwartzing & Wöhr, 2011). Adult rats who have experienced stress during adolescence or those with high trait anxiety will emit more anxious USVs (Yee et al, 2012; Schwartzing & Wöhr, 2011).

Alternatively, 50kHz USVs in adults signal anticipatory pleasure (Wöhr et al., 2008).

Anticipatory pleasure was assessed immediately prior to the SAT to outline any differences in both positive and negative emotionality in rats from the two breeding schemes.

Methods

Animals

Sixteen Wistar (8 female) breeding animals were used in 4 intermittent and 4 continuous breeding pairs. Animals were housed in Makrolon® polycarbonate cages (356x485x218mm; ACR Optirat), in temperature ($21 \pm 2^\circ\text{C}$) and humidity controlled (55-60%) environment, with reversed 12 hour light/dark cycles (lights on at 1900 hours). Access to food and water was ad libitum. All procedures were approved by the Animal Ethics Committee of Victoria University Wellington (AEC 25843). Continuous breeding pairs were housed together for the entirety of experiment. Intermittent breeding pairs were housed together for two weeks at a time before the male was separated to ensure no contact with pups. Day of birth for the pups was considered PND0. All litters were housed with the dam (and sire in continuous pairs) until weaning at PND21. Intermittent males were placed back into the females cage immediately after weaning. Following weaning, pups were housed with sex-matched litter mates for the rest of the experiment. Pups were weighed on all days they were tested plus PND14 and PND45. Apart from that, handling was limited to the experiments, as it is known to affect parental care and offspring's stress reactivity in adulthood (Meaney, 2001). Preweaning deaths were also recorded.

Materials/ Experimental Setting

All experiments were carried out in the same room, under white light and temperature ($21 \pm 2^\circ\text{C}$) and humidity controlled conditions (55-60%). All video recording were made using EthoVision software (EthoVision XT 9; Noldus Information Technology, Leesburg, VA) and audio recordings were made with an UltraMic (205K, 16 bi; Java Sound) and Raven Pro Software (Cornell University, Ithaca, NY). The video camera was attached to the ceiling to allow a bird's eye view of experiments.

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

All statistical analysis was performed using SPSS (IBM SPSS Statistics for Windows, Version 23.0. Armonk, NY: IBM Corp.).

USV potentiation

USV potentiation was measured on PND4 and PND10. Before the experiment both parents were removed from the homecage and placed in a separate cage or in a black (25x45x30cm) experimental box, with a small amount of bedding, for a 10 minute habituation period. One pup was then removed from the homecage and placed in a 10cm round (15cm high) recording chamber, with a small amount of bedding at the bottom. The pups USVs were recorded for 5 minutes. After this first isolation period (iso1) the pup was placed in the experimental box with either its mother or father. If the reunion type was with the litter then the pup was placed back in the homecage with its littermates. The reunion period lasted 5 minutes and, in the case of mother and father reunions, video was recorded.. After the reunion, the same pup was placed back in the chamber for the second isolation period (iso2) where USVs were again recorded. From each litter, 2 pups underwent trials with their mother, 2 with their father and 2 with the litter mates for a total of 6 trials per litter on each day of testing. Videos were manually scored using EthoVision for licking, grooming, huddling and retrieving behaviours by the parent. Sound files were analysed using Raven where a filter was applied at 30-50kHz in order to count individual calls.

Father Interest in pups

To assess the bonding of sires to pups, father interest experiments were carried out on PND5 and PND9. Sires were placed in a 50x50x25 black experimental box (Figure 1) with no bedding for 10 minutes of habituation. In an adjacent (25x20x25cm) chamber were two ramekins that the pups would be placed in later. The pup and sire chambers were separated by a wire separator so that the sire could see, smell and hear the pups but not touch them.

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

After 10 minutes of habituation, 2 pups were placed in each ramekin in the pup chamber for an additional 6 minutes. Video and USVs were recorded throughout the experiment.

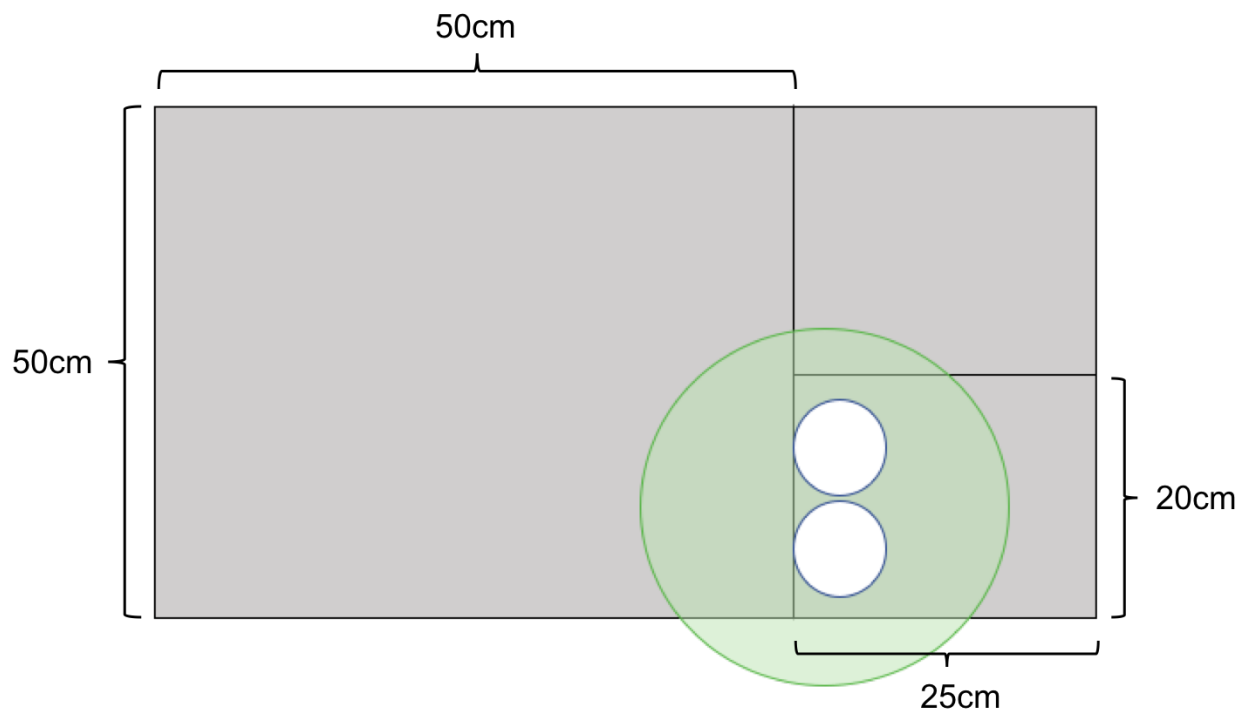


Figure 1: Birds eye view of experimental box for father interest paradigm. The green circle (superimposed with EthoVision after recordings were made) indicates the “pups” area. Cumulative duration of the sires nose in this “pup” area was recorded as a measure of father interest in the pups.

Novelty suppressed feeding (NSF)

NSF was carried out between PND40 and PND45 as a measure of anxiety-like behaviour in adolescent pups. Higher maternal care has previously been found to reduce anxiety-like behaviour in this paradigm (Caldji et al., 1998). Animals were given fruit loops (half each) for 3 days prior to experiment onset as rats can be wary of unfamiliar foods (Barnett, 2009). On the day of experiment, half a fruit loop was placed in the centre of a 50x50x25cm black experimental box. Animals were placed at the edge of the open field and recorded for 10 minutes. Latency to retrieve the fruit loop was manually recorded and cumulative duration spent around the perimeter and in the centre of the field was recorded. The dimensions of the center field (26 x 26 cm) were created in Ethovision.

Successive Alleys Test (SAT)

Higher maternal care leads to lower anxiety-like behaviour in the elevated plus maze (EPM; Priebe et al., 2005) and open field tests (Caldji et al., 1998). The SAT is a more sensitive measure of anxiety than the EPM (Deacon, 2013) as it incorporates four levels of alleys that are increasingly anxiety-inducing as the wall height and floor width decrease from alley 1 through to alley 4 (for more information refer to appendix A). However, as with the EPM, the standard SAT suffers from the one trial tolerance phenomenon and does not allow repeated assessment of anxious behaviour. For that reason, we recently developed an adapted SAT (mSAT), in which small rewards (fruit loops) were placed in the arms, to entice the animals to venture out into the open arms (Kidwell, 2018, Master Thesis). Animals were placed in the mSAT (Figure 2) so they were facing the back wall of alley 1. The mSAT was performed 10 times on alternating days, with an 11th trial occurring 3 weeks after the 10th trial, to assess potential long-term changes in anxiety-like behaviour. Birds eye view video was recorded for a duration of 5 minutes. Latency to retrieve fruit loops and the alley that

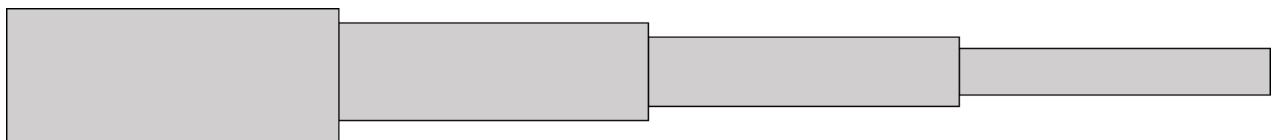
DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

they were consumed in was recorded manually. SAT was cleaned with disinfectant between animals and any remaining fruit loops were replaced.

Anticipatory Pleasure (AP)

To investigate differences in emotional state, an AP paradigm was carried out, from PND90, in order to measure anxious (negative) and anticipatory (positive) calling. AP was carried out immediately before the mSAT to measure the animal's anticipation of having access to fruit loops in the mSAT. Animals were placed in an experimental box (25x45x30cm) with a small amount of bedding at the bottom for 5 minutes. Sound recordings were made using Raven Pro Software. Negative affect calls were operationalised as vocalisations between 18-32kHz and positive affect calls as those between 35-70kHz (Schwartz & Wöhr, 2011). Video recordings measured anticipatory locomotion. Like the mSAT, this was repeated for 10 trials on alternating days with an 11th trial occurring 3 weeks following the 10th.

(a.)



(b.)

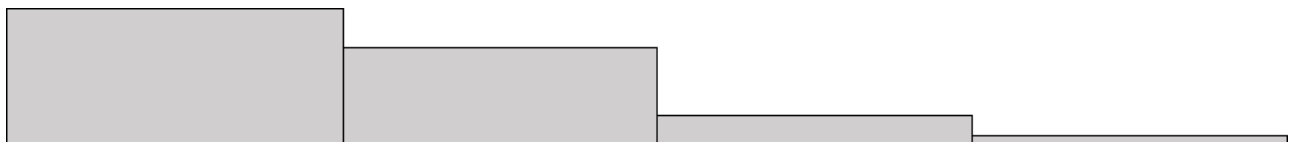


Figure 2 Birds eye and side view of SAT. For each figure Alley 1 is on the far left (a.) shows the SAT from above and illustrates the decrease in floor width with each alley, (b.) shows the alley from the side and demonstrates the decreasing wall height

Results

Potentiation of USVs: Ratio of calls made in iso2 compared to iso1 was calculated by dividing number of calls in iso2 by number of calls in iso1 for each pup. Checks for normality found that this data set was positively skewed so a log transformation was applied. This was subjected to a 2 (Group; Continuous, Intermittent) x 2(Age; PND4, PND10) x 3(Reunion; Mother, Father, Litter) between subjects analysis of variance (ANOVA). Levene's Test of Equality of Variances was significant so the Greenhouse-Geisser correction was used. A significant main effect of Reunion was found [$F(2,134) = 9.088, p < 0.001$]. Pairwise comparisons found that calling rate was significantly reduced after reunification with sires compared to with dams or littermates (*figure 3*). A significant interaction between Reunion type and Group was found [$F(2, 134) = 10.496, p < 0.001$]]. Pups bred intermittently showed potentiation of USVs after the reunion phase in all conditions except reunion with sires. Average USV calls across both isolation periods were compared between intermittent and continuous groups. However, no significant differences were found.

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

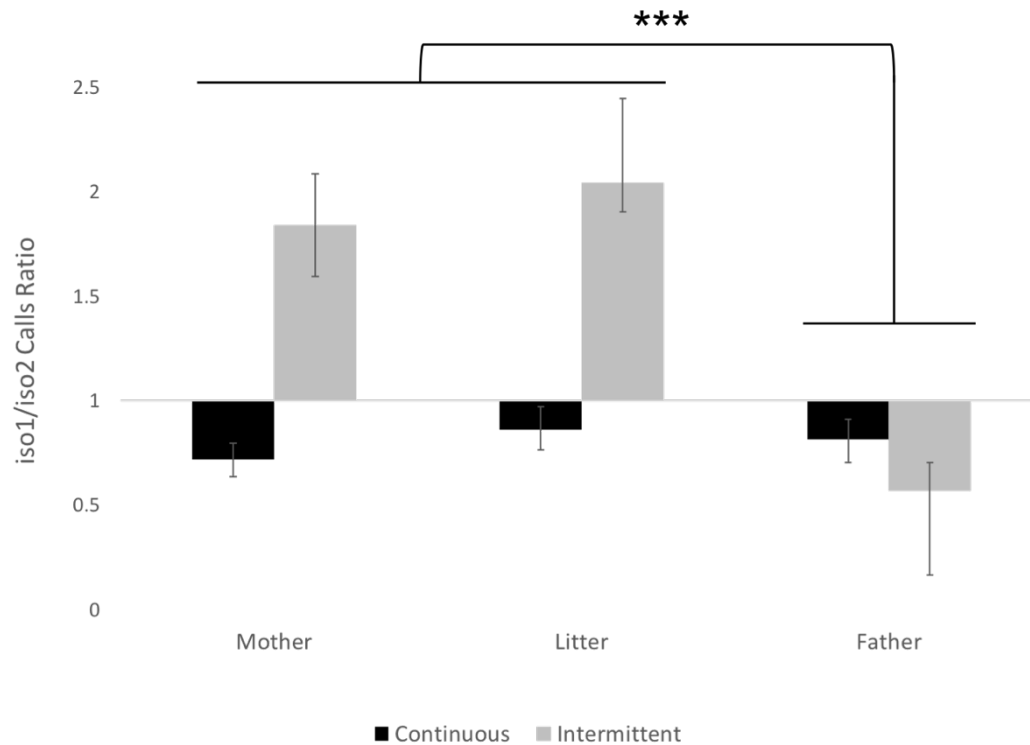


Figure 3: Ratio of calls made by pups in iso2 compared to iso1. Data is presented as means $\pm$ SEM. The 3 reunion conditions (Mother, Father, Litter) are presented along the x-axis with grey bars indicating mean calling scores from intermittent pups and black bars indicating mean calling scores from continuous pups. Scores larger than 1 indicate an increase in calling rate following reunification period (potentiation) and scores less than 1 indicate a decrease in calling rate following reunification (depotentiation). A significant main effect of reunion type ($p < .001$) and a significant interaction between group $\times$ reunion type ($p < .001$) was found.

Pup Development. Weaning weights (PND21) underwent a 2(Sex; Male, Female) by 2(Group; Intermittent, Continuous) between subjects ANOVA. No significant effect of sex was found but a significant main effect of Group [$F(1,315) = 8.79, p = 0.003$] indicated that continuous pups were heavier at weaning than intermittent pups, as seen in *figure 4*.

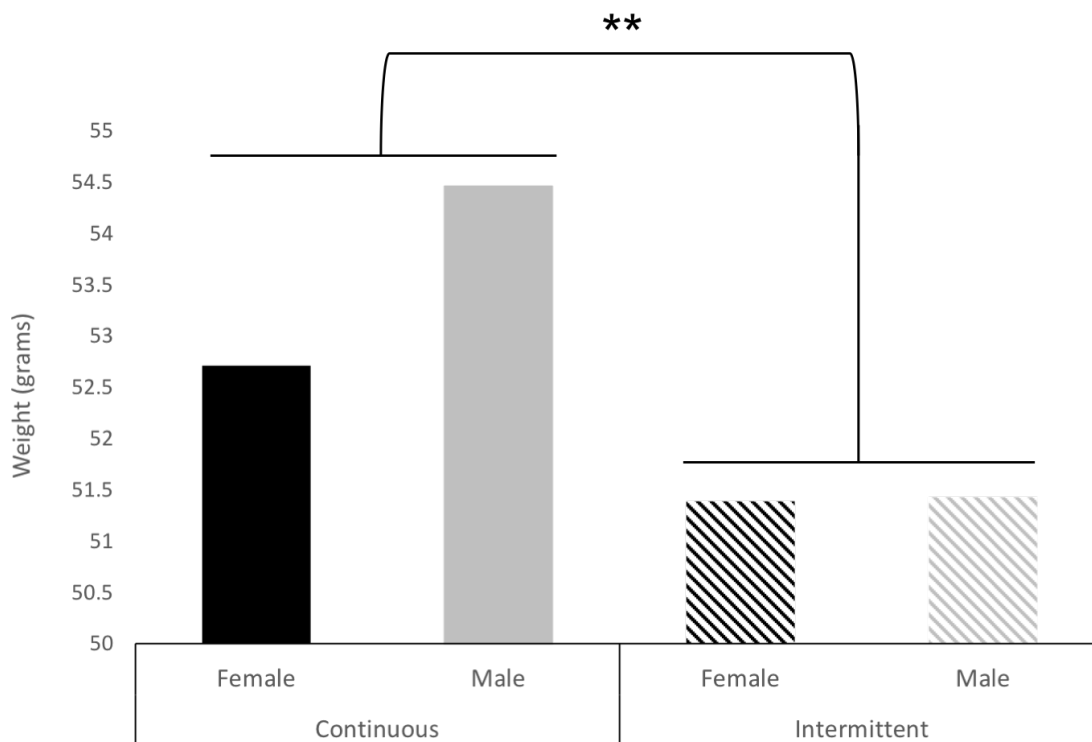


Figure 4: Average weight (grams) at weaning (PND21). Data is presented as means $\pm$ SEM with grey and black bars representing male and females means respectively. Striped bars indicate scores from intermittent animals and solid bars represent continuous animals. The only significant effect found was of groups ($p=.003$), indicating that continuously bred animals were heavier at weaning on average.

Father Interest. Number of seconds that the father's nose spent facing the pups (within 10cm) were compared between continuous and intermittent fathers using an independent samples t-test (*figure 5*). Levene's Test of Equality of Variances was significant therefore equality of variances was not assumed. The difference between groups was non-significant ($p = 0.505$).

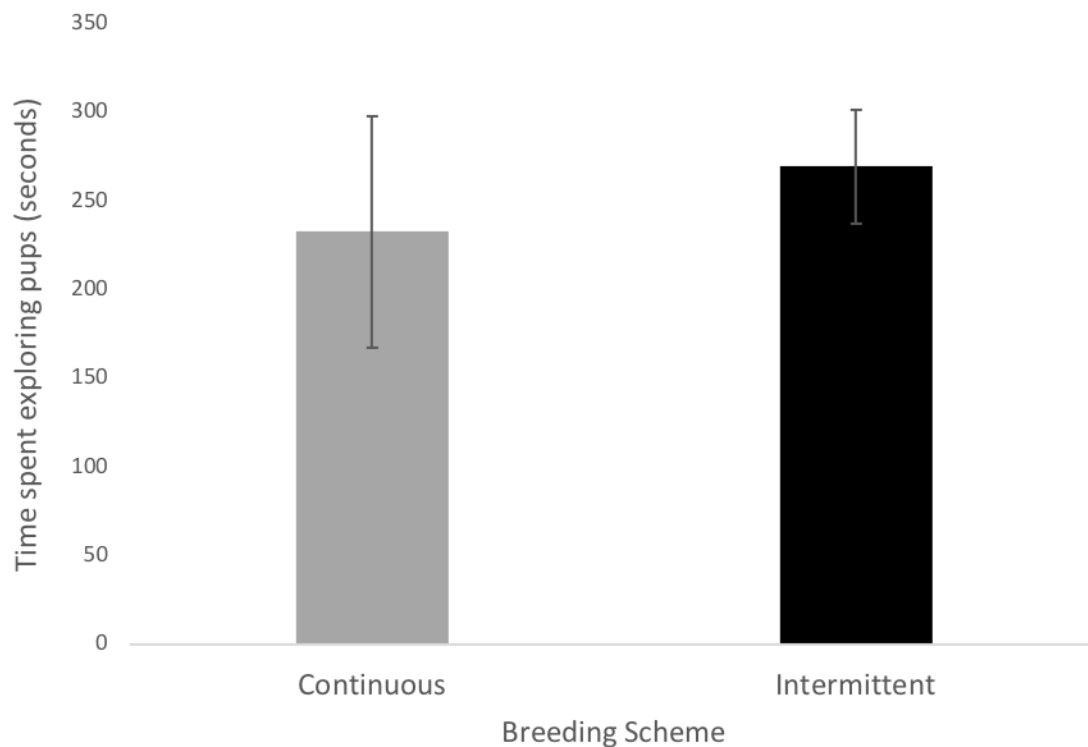


Figure 5: Time (seconds) spent exploring pups in father interest experiment. Data is presented as means $\pm$ SEM. No significant differences between the two breeding schemes was found

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

NSF. Latency to retrieve food reward was recorded in seconds. If the animal did not retrieve the food reward during the 10 minute experiment then they were given a latency of 600 seconds. This was subjected to a 2 (Group; Continuous, Intermittent) by 2 (Sex; Male, Female) between subjects ANOVA which revealed a significant main effect of group [$F(1, 56) = 15.178, p < 0.001$]. Continuously bred animals retrieved the fruit loop faster than intermittently bred animals (*figure 6*). Percentage time spent in the center of the testing apparatus was subjected to the same ANOVA and again found a significant effect of group [$F(1, 65) = 4.336, p = 0.041$]. Continuously bred animals spend more time in the center than intermittently bred groups (*figure 7*).

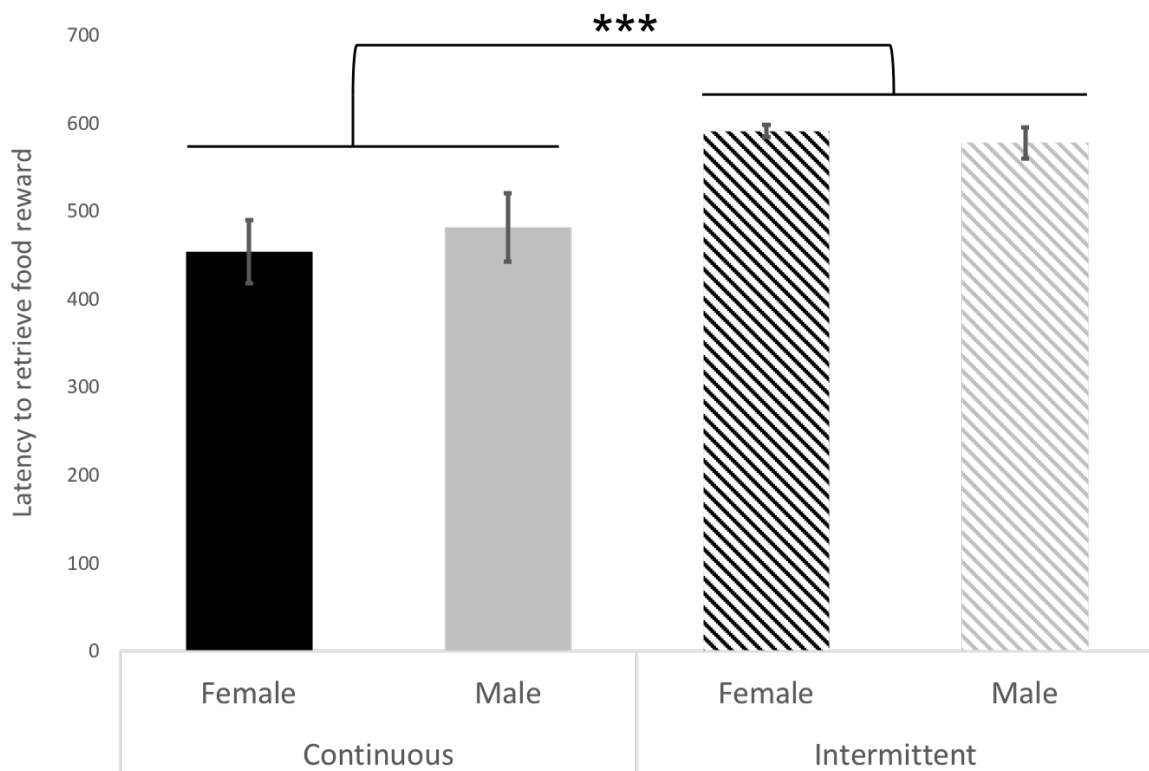


Figure 6: Average latency to retrieve food reward in NSF paradigm. Data is presented as mean $\pm$ SEM, with sex and breeding scheme depicted along the x-axis and average latency (seconds) along the y-axis. A significant main effect of breeding scheme was uncovered ($p < .001$), as continuously bred offspring retrieved the food reward faster than intermittently bred offspring.

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

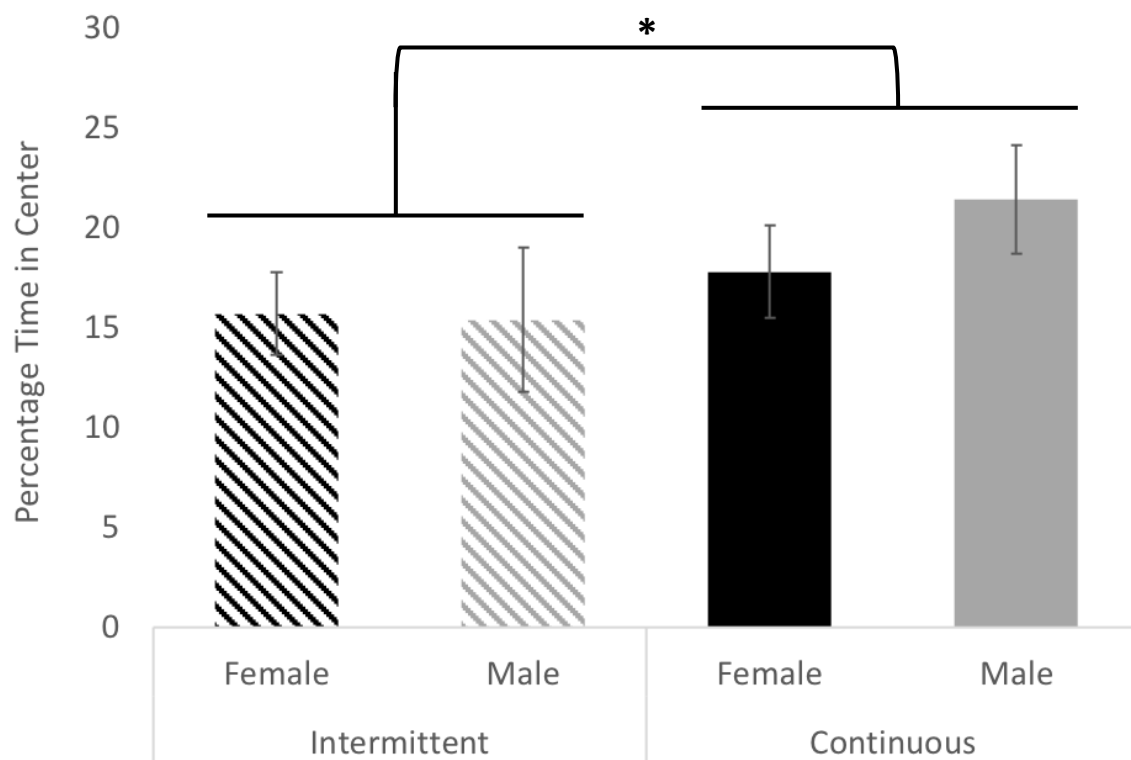


Figure 7: Percentage time in center of arena in novelty-suppressed feeding paradigm. Data is presented as means $\pm$ SEM, with condition (sex and breeding scheme) depicted along the x-axis and percentage of time (seconds) spent in the center of the arena depicted on the y-axis. A marginally significant effect of breeding scheme was uncovered ($p=.041$).

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

SAT. Latency to enter and cumulative duration data were very skewed due to most males staying in alley 1 for the majority of the days tested. Alleys 3 and 4 scores were combined as these were the most anxiety-inducing alleys as both had walls lower than the height of the rat. Both sets of data then underwent logarithmic transformations to achieve normality. Both latency to enter and cumulative duration scores underwent a mixed ANOVA with Group and Sex as between subjects factors and day as the within subjects factor.

For cumulative duration, Mauchly's Test of Sphericity and Levene's Test of Equality of Variances were both significant indicating unequal variances across all within and between subjects factors. Therefore, the Greenhouse-Geisser correction was applied to all statistics. A significant main effect of day was found [$F(5.662, 305.726) = 18.988, p < 0.001$], with animals spending more time in alleys 3 and 4 in the 10th trial compared to the 1st across all conditions (*figure 8*). Significant interactions were found for Day x Sex [$F(5.662, 305.726) = 6.851, p < 0.001$], Day x Group [$F(6.472, 305.726) = 6.472, p < 0.001$], and Day x Sex x Group [$F(5.662, 305.726) = 3.679, p = 0.02$]. Pairwise comparisons found that cumulative duration in alleys 3 and 4 did not significantly differ between the 10th trial and the 11th trial 21 days later (*figure 9*). The cumulative duration in alleys 3 and 4 across the 11 trials did not significantly change in male rats, with no significant difference between days 1 and 10. Therefore, the Day x Sex interaction was driven by the females spending more time in alleys 3 and 4 during later trials. Inspection of *figure 8* does not give a clear indication what drove the Day x Sex x Group interaction.

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

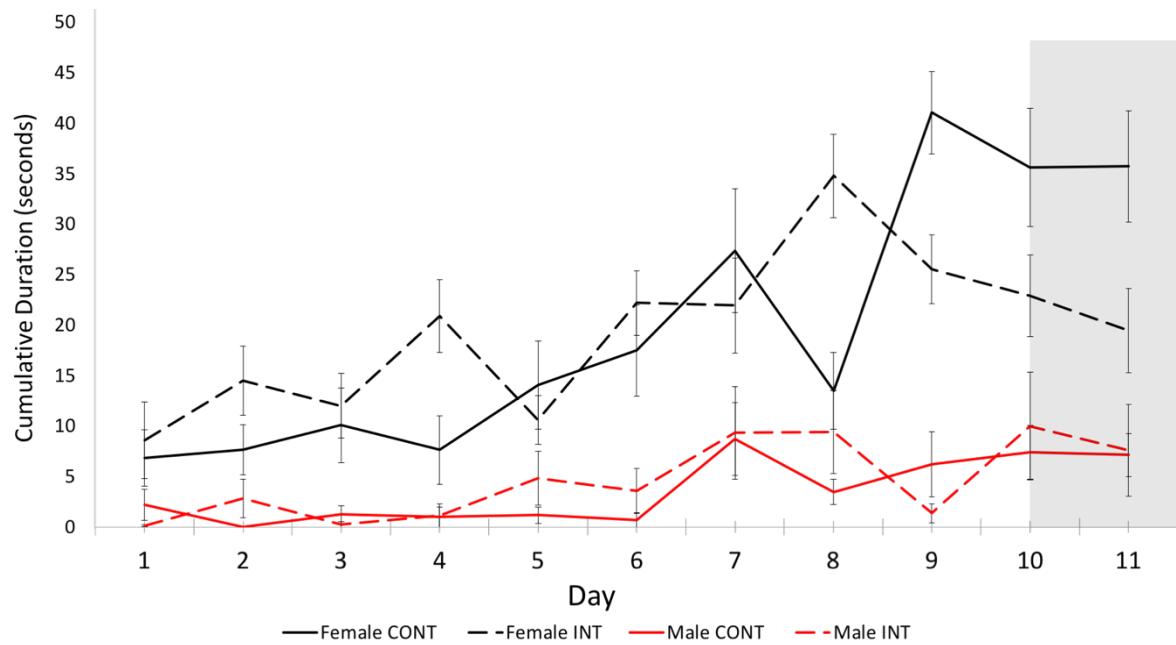


Figure 8: Cumulative duration in alleys 3 and 4 of the SAT. Data is presented as mean $\pm$ SEM for each of the four sex x groups conditions. Days of testing are indicated along the x-axis with trials 1 to 10 occurring on alternating days and the 11th trial occurring 21 days after the 10th. Dashed lines are average scores from intermittently bred rats and solid are scores from continuously bred rats. Black lines are female scores and red lines are males. A significant main effect of day ($p < .001$) and significant interaction between day x sex ($p < .001$), day x group ($p < .001$) and day x sex x group ($p = .02$) were uncovered.

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

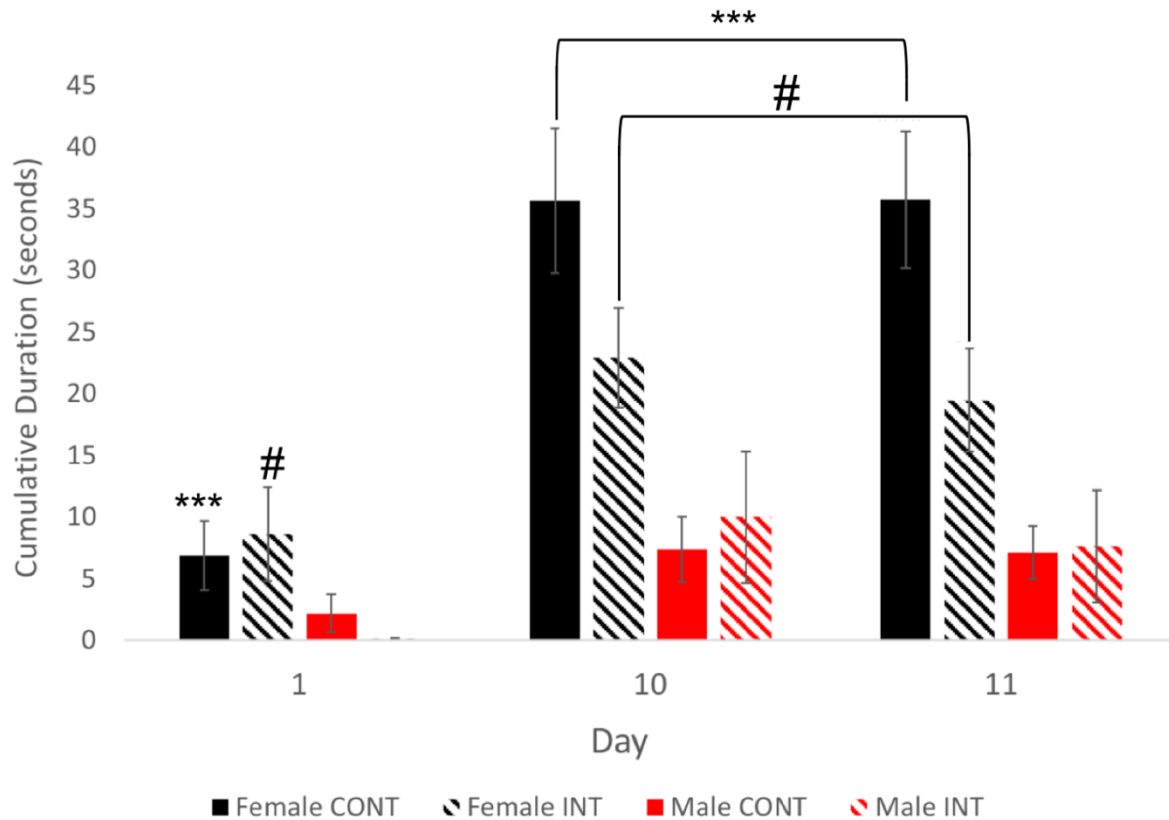


Figure 9: Comparison of cumulative duration (CD) in alleys 3 and 4 of SAT on days 1, 10, and 11 of testing. Data is presented as mean $\pm$ SEM CD (seconds) for each of the four sex x group conditions. Black and red bars portray scores from males and females respectively, whilst solid bars indicate scores from the continuous animals and striped bars are scores from intermittent animals. CD did not significantly differ between days 1, 10, 11 for all male rats. Intermittent and continuous females CD on days 10 and 11 both significantly differed from their respective CD on day 1. There was no significant difference CD on days 10 and 11 in both intermittent and continuous females.

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

Average latency to enter alley 3 and 4 is depicted in *figure 10*. If the animal failed to enter an alley then a latency of 300 seconds was given. The mixed ANOVA again found significant main effects of Day [$F(4.991, 269.526) = 12.886, p < 0.001$], and Sex [$F(1, 54) = 67.993, p < 0.001$]. Overall latency to enter alleys decreased with cumulative trials but females had lower latencies than males on each day of testing. A significant interaction between Day and Sex [$F(4.991, 269.526) = 67.993, p < 0.001$] was found. Post hoc analysis found that males latency scores did not significantly differ between days 1, 10 and 11 whereas females latencies were significantly higher on day 1 than on days 10 and 11 (depicted in *figure 11*). As with cumulative duration data, females latency to enter alleys 3 and 4 did not significantly differ between day 10 and day 11, 3 weeks later.

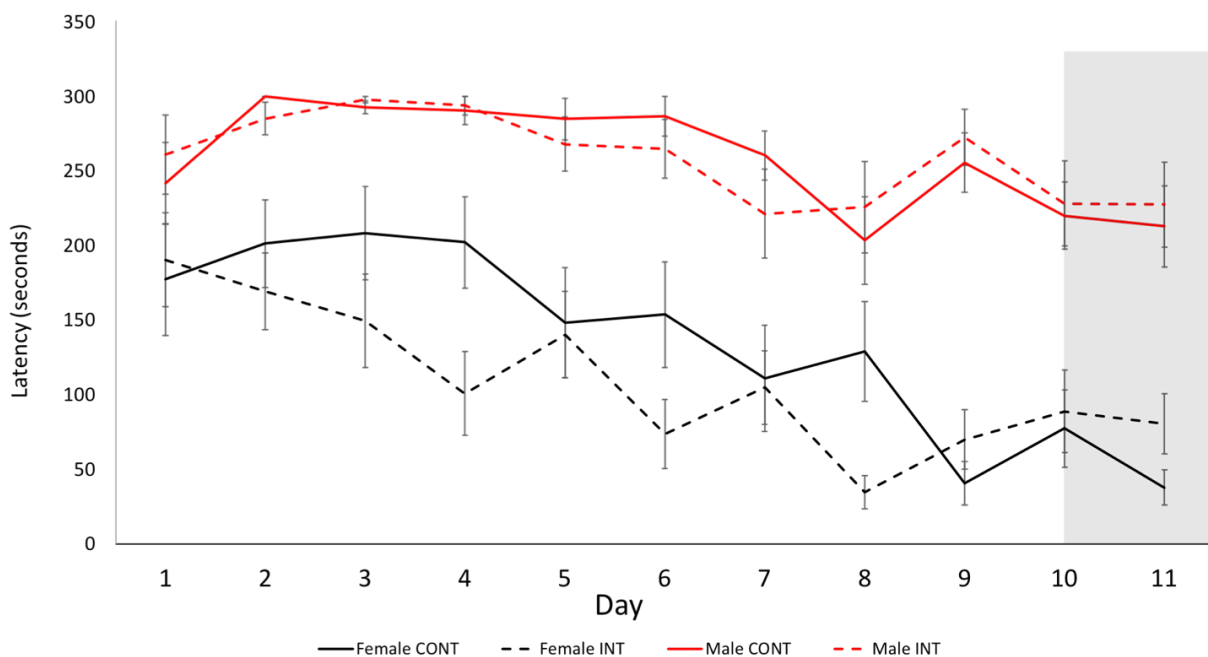


Figure 10: Average latency to enter alleys 3 and 4 of SAT on trials 1 to 11. Data is presented as means $\pm$ SEM for each of the four sex x group conditions. Days of testing are indicated along the x-axis with trials 1 to 10 occurring on alternative days and the 11th trial occurring 21 days after the 10th. Dashed lines are average scores from intermittently bred rats and solid lines are scores from continuously bred rats. Black lines are female scores and red lines are male. Significant effects of day ($p < .001$), sex ($p < .001$) and a significant day x sex interaction ($p < .001$) were found.

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

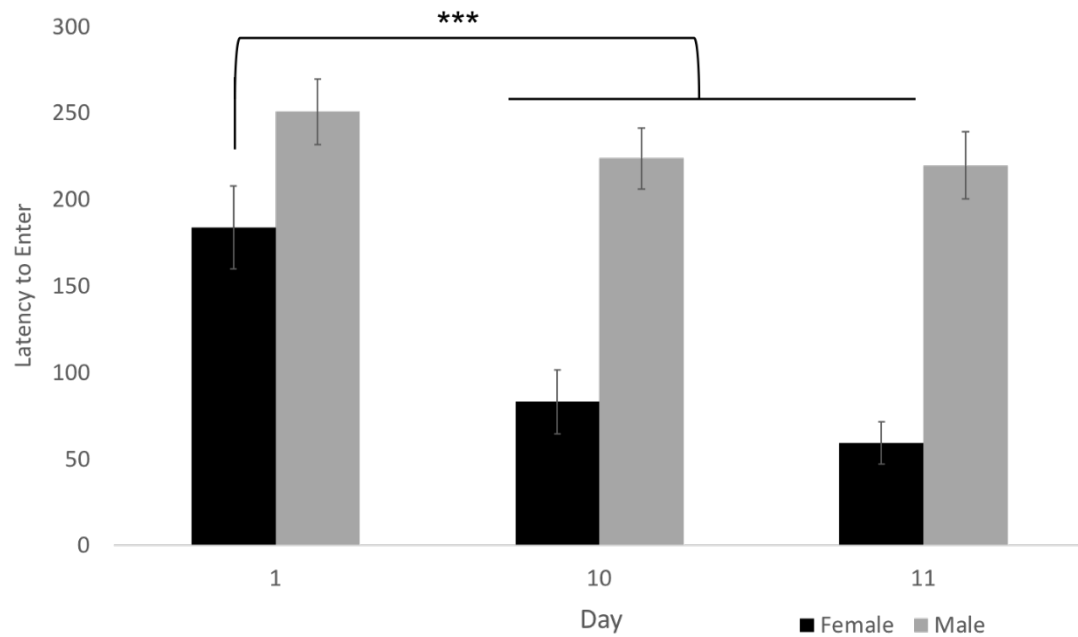


Figure 11: Comparison of mean $\pm$ SEM of latency to enter alleys 3 and 4 of SAT on days 1, 10, and 11 of testing. Days are presented along the x-axis. Black and grey bars portray scores from females and males respectively. No significant differences in latency across the 3 days was found in males. In females, latency on days 10 and 11 were both significantly reduced compared to day 1. No significant difference between latency on day 10 and 11 was found in females

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

Anticipatory Pleasure. 50kHz calls were operationalised as calls falling in the range of 35-70kHz, and 22kHz calls were those between 18 and 32kHz. Consistent with past research (Schwartz & Wohr, 2011), there was a high level of individual variability in number of calls between animals. Therefore, calling rate on days 2 through to 11 were calculated as the percentage change from calls on day 1. The number of 22kHz calls were positively skewed, therefore a log transformation was applied in order to achieve normality. A 2 (Group; Intermittent, Continuous) by 2 (Sex; Female, Male) by 10 (day) mixed repeated measures ANOVA failed to find any significant effects (*figure 12*).

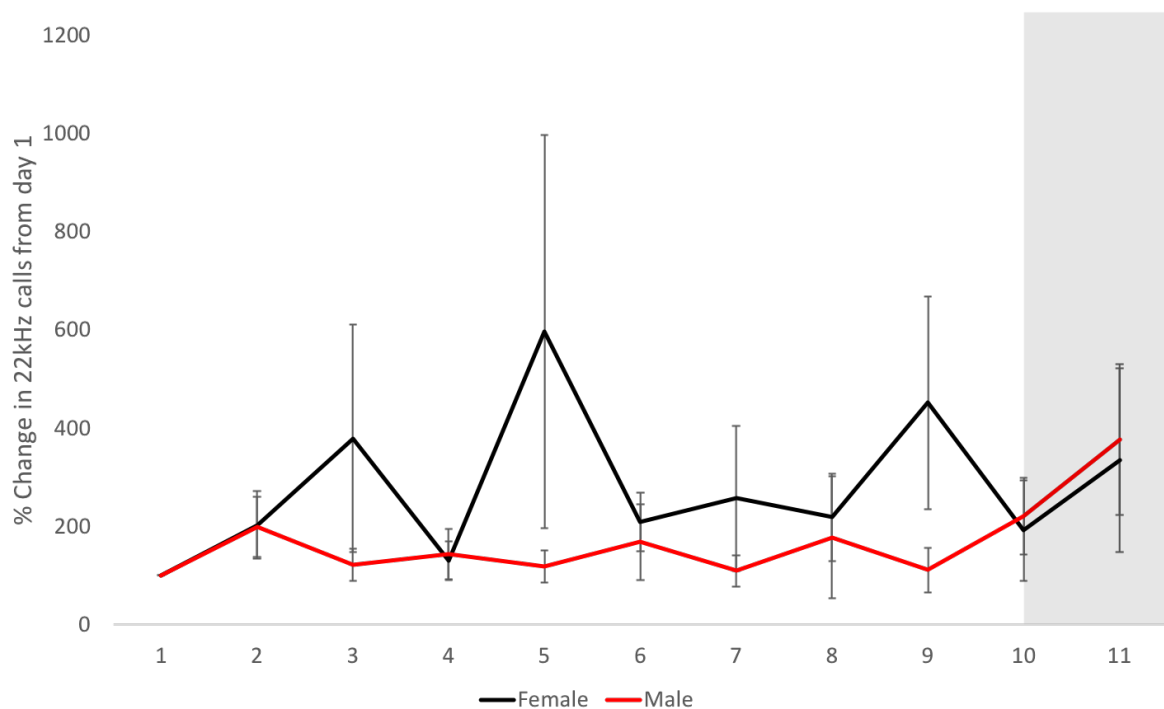


Figure 12: Percentage change in 22kHz calls from day 1 in anticipatory pleasure paradigm. Data is presented as means $\pm$ SEM with days depicted along the x-axis and percentage change in calls (compared to day 1) depicted along the y-axis. All trials were one days apart with the exception of trial 11 that occurred 21 days after trial 10. No significant main or interaction effects were uncovered

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

50kHz calls underwent the same ANOVA. Mauchly's test of sphericity was found to be significant so Greenhouse-Geisser corrections were applied. A significant main effect of day was found [$F(5.129, 128.218) = 3.284, p = 0.007$]. Post hoc pairwise comparisons found marginally significant differences in calling rate between day 1 and days 10 and 11 ($p=.049$) and no change in calling rate between day 10 and day 11 (*figure 13*).

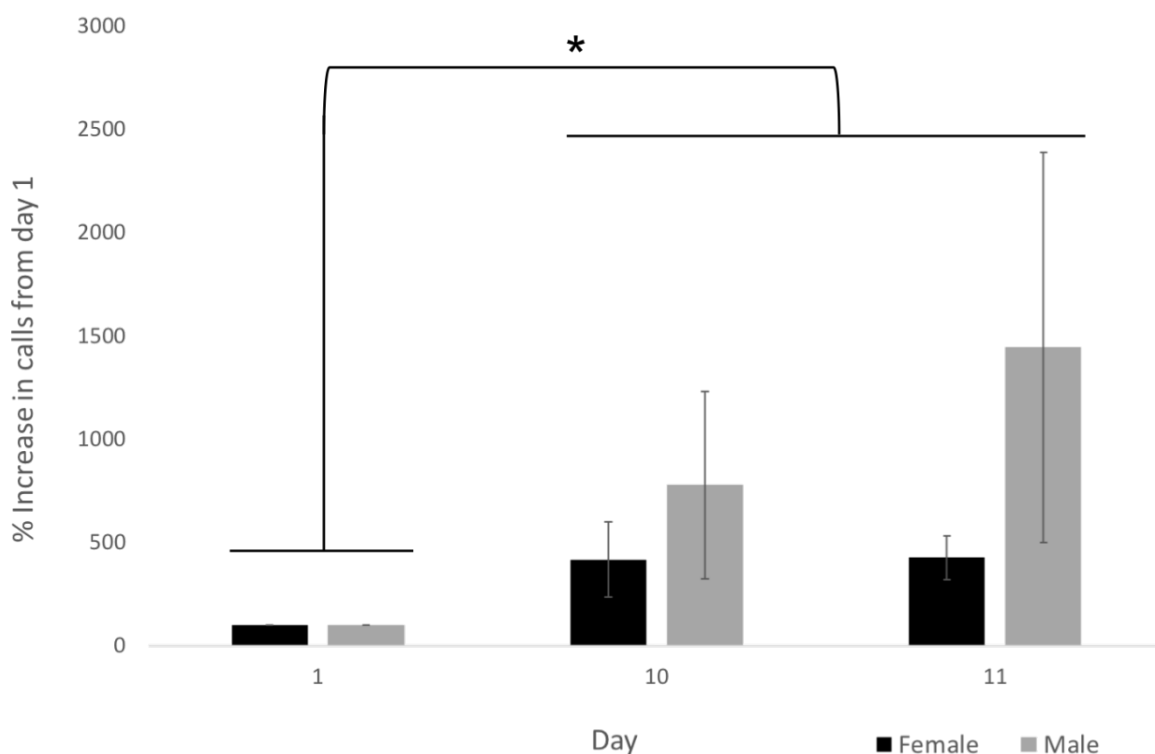


Figure 13: Percentage increase in 50kHz calls in days 10 and 11 compared to day 1 in anticipatory pleasure paradigm. Data is presented as means $\pm$ SEM, with days presented along the x-axis. Black and grey bars represent means from females and males respectively. The only significant effect found was of day ($p=.007$), with more 50kHz calls on days 10 and 11 than on day 1.

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

Locomotor activity measured during the anticipatory pleasure phase was measured by distance moved (cm) in the 5 minute period. This data underwent a mixed ANOVA with Group and Sex as between subjects factors and day (1 through to 10) as the within subjects factor. No significant main effects of, or interaction effects including, day were found. Data was then averaged across days and subjected to a 2(group) by 2(sex) between subjects ANOVA (*figure 14*). Significant main effects of both Sex [$F(1, 315) = 178.383, p < 0.001$] and Group [$F(1,315) = 5.074, p = 0.025$] were found. Overall females and continuous rats moved a greater distance than males and intermittent rats respectively.

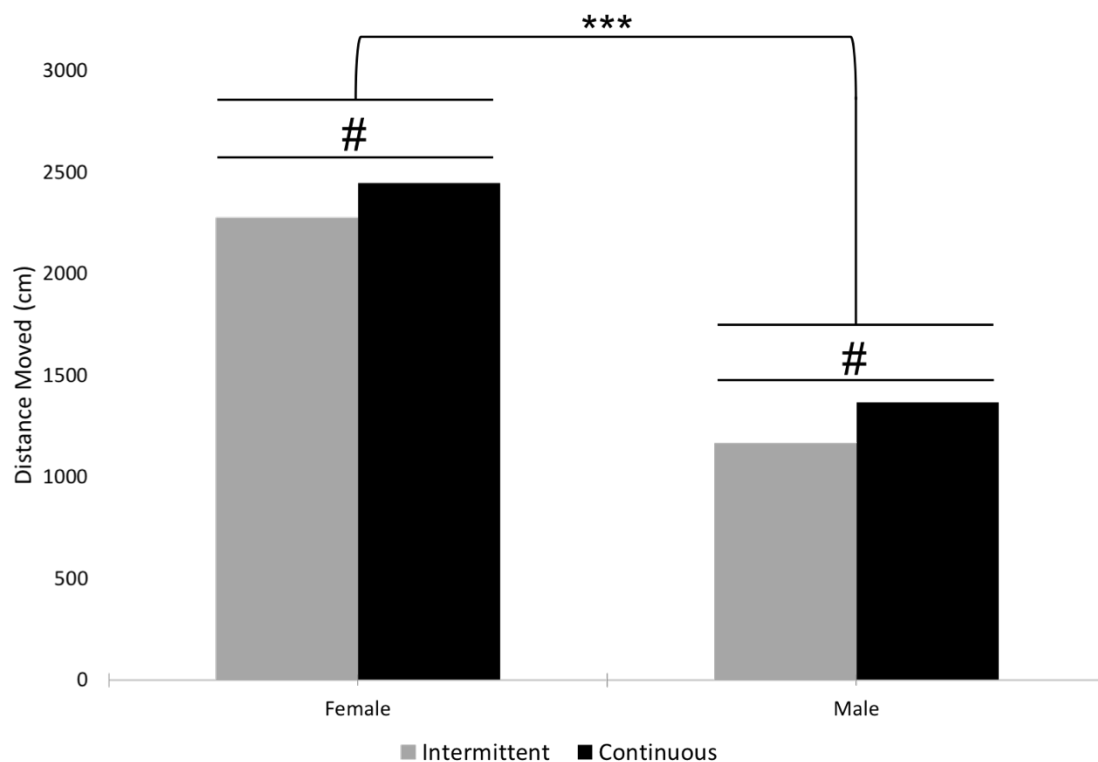


Figure 14: Average distance moved (cm) across 11 days of anticipatory pleasure testing. Data is presented as means $\pm$ SEM, with black and grey bars representing scores from continuous and intermittent rats respectively. Significant main effects of both sex ($p < .001$) and group ($p < .025$) were found.

Discussion

This study examined the influence of paternal care on offspring's emotional behaviour by comparing intermittent and continuous breeding schemes. First, the bond between parent and offspring was measured in two experiments. Infant attachment to parents was measured by infant USVs which also served as an indirect measure of the parental care (through LG) they received. Father attachment to pups was measured by a father-interest paradigm. Second, offspring's emotional behaviour was measured in adolescence with NSF, and adulthood using mSAT and anticipatory pleasure paradigms.

Infant and Parent Attachment. Pups exhibited USV potentiation after reunification with the dam or littermates and depotentiation after reunification with sires, in the parental separation paradigm. There was no difference between mother and litter reunions on calling rate. This appears inconsistent with literature stating that potentiation occurs only after reunification with dams and not littermates (Shair et al., 2008b). However, USVs are emitted following a change in several stimuli including body temperature, contact with adults or littermates, and smell (Ehret, 2005). Pups' calling rate depends on the integrated value of these stimuli. The littermate reunification period occurred within the home-cage where the pups were exposed to warmth from littermates and familiar smells. Therefore, the combination of stimuli that the pups were exposed to in this period were enough to elicit a similar level of USV potentiation as produced after reunification with dams.

A significant interaction between reunion type and breeding scheme revealed that the effect of reunion type was driven by the intermittently bred offspring, as continuous litters produced depotentiation of USVs in all conditions. This suggests continuously bred pups are not as dependent on dams (and littermates) as intermittently bred pups. No difference in average USV rates suggest that the overall level of LG received by pups in the two breeding schemes was the same, as level of LG received by pups in early development correlates

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

positively with USV rate (Wohr & Schwarting, 2008). Dewsbury (1985) found that sires can contribute to LG so continuous dams may have performed LG less than intermittent dams. Therefore, continuous pups could have experienced more periods of isolation from the dam (compared to intermittent), which could have made subsequent isolation in the USV experiment less distressing. USV potentiation research is generally carried out in intermittently bred animals and our data from intermittent pups is consistent with previous research (Shair, Muller, & Moore, 2008b; Hofer, 1996). Thus, the interaction between reunion type and breeding scheme discovered here warrants further investigation.

Continuously bred pups were also heavier at weaning, which is consistent with previous research (Allen et al., 2013). Dennenbery and Whimby (1963) found that pups that were handled during their infancy had a larger weaning weight and displayed less anxiety in an open field than unhandled pups. It is now understood that the effect of handling is due to an increased level of maternal LG (Francis et al., 1999). Therefore, the higher weaning weight of continuously bred pups is consistent with their behaviour in parental separation and NSF experiment. Together this data suggests that continuously bred pups receive better parental care and perhaps more LG than intermittently bred pups.

Offspring Emotional Behaviour in Adolescence and Adulthood. The NSF measured conflict anxiety in adolescent offspring and found a significant effect of breeding scheme. Continuously bred offspring exhibited lower anxiety-like behaviour than intermittently bred offspring, by spending relatively more time in the center of the apparatus and retrieving the food reward faster.

USVs were measured in an anticipatory pleasure paradigm as an indicator of emotional state (Wohr et al., 2008). Consistent with previous work, there were high levels of individual variability in USVs (Schwarting & Wohr, 2011). No significant effects of day, sex or breeding scheme were found on number of aversive (22kHz) calls made. Wistars tend to

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

emit fewer 22kHz calls compared to other strains (Walker, Naicker, Hinwood, Dunn, & Day, 2009). However, the current results are likely because the testing environment was not aversive. The offspring may be more likely to emit 22kHz calls had there been no previous exposure to the experimenter and habituation in previous experiments.

A significant increase in 50kHz calls over course of 10 days across all conditions was found. As anticipatory calls (50kHz) measure positive emotional states (Brudzynski, 2005), it suggests that all animals had an increase in positive state throughout the experiment. Interestingly, there was no significant reduction in calling between the 10th and 11th trial, despite a 21 day gap between them, suggesting that emotional reactions to the anticipatory pleasure and mSAT paradigms remain stable for at least 21 days.

Anticipatory locomotor activity was recorded as another indicator of emotional state. The experiment uncovered significant effects of sex and breeding scheme but not day. Increased locomotor activity in female, compared to male, rats is a well-documented effect, and can underlie differences in anxiety-like behaviour (Donner & Lowry, 2013; Fernandez, Gonzalez, Wilson, & File, 1999). A surprising finding was that continuously bred animals had increased anticipatory locomotor activity. Increased locomotor activity can correlate with reduced anxiety (Graham, Yoon, Yee, & Kim, 2009), suggesting that continuously bred offspring were less anxious than intermittently bred offspring.

Cumulative duration and latency to enter data from the mSAT both produced significant effects of sex and interactions between day and sex. No main effect of breeding schemes was uncovered. Female rats exhibited lower anxiety-like behaviour than males, and females anxiety-like behaviour reduced over period of 10 days while males remained stable. This novel finding for the mSAT is consistent with other rodent models of anxiety (Donner & Lowry, 2013). Several studies report sex differences in the development of anxiety following stress exposure in rodents (Donner & Lowry, 2013; Pryce & Feldon, 2003). Male rats

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

exposed to early life stress show increase anxiety behaviour in EPM or open field but females do not (Prusator & Meerveld, 2015). However, there was no interaction between sex and breeding scheme, so sex differences here are not due to different reactions to stress caused by the breeding conditions. Male rats may require more motivation to explore the alleys than females do, as previous work in our lab found no sex differences in the mSAT when all animals were food deprived (Kidwell, 2018, Master thesis).

Our lab developed the mSAT as a solution to the one-trial tolerance (OTT) problem. OTT occurs in the EPM, where previous exposure to the apparatus reduces the effect of anxiolytics (Carobrez & Bertoglio, 2005), which severely reduces the retest reliability of the EPM. The current study found that male rats did not significantly increase their time spent in the more anxiogenic alleys (3 and 4) over the course of the experiment. Suggesting that the mSAT is not subject to OTT in male rats. Females did significantly increase their time in alleys 3 and 4 over the course of the experiment. However, the increase was small (peaking at about 30 seconds out of a total of 300) and appeared to plateau towards the end of the experiment. Therefore, the mSAT could be a more reliable measure of conflict anxiety in rats than the original SAT or the EPM.

General Discussion. Continuous breeding schemes had a positive effect on offspring's emotional behaviour in infancy and adolescence. Continuously bred pups had patterns of USV potentiation that suggested less dependence on dams, less anxiety-like behaviour in NSF and were heavier at weaning. Continuously bred pups also had higher locomotor activity in adulthood, which could indicate lower anxiety. However, adult emotional behaviour was more strongly determined by sex than by breeding scheme. There was no relationship between pup USVs and adult anxiety as seen in previous work (Brunelli, 2005; Insel & Hill, 1987). However, this relationship has only been found in selectively bred

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

animals (Wohr & Schwarting, 2008), so it is not surprising that it was not present in the wild-type rats used here.

Continuous breeding schemes were more beneficial to pups although the effect did not last into adulthood. Parental care (through LG) can alter offspring's stress reactivity through epigenetic alterations to the HPA axis (Francis et al., 1999). The post weaning environment can in turn alter the epigenetic changes due to the preweaning environment. For example, Francis et al. (2002) found that post weaning enrichment could attenuate the heightened stress reactivity in offspring due to maternal separation in infancy. Therefore, the differences in infant and adolescent behaviour in the two breeding schemes could have diminished over the course of early adulthood as all rats had identical post-weaning environments. Which resulted in no significant effects of breeding scheme in adult measures of emotional behaviour.

Maternal care provides feedback from stressors in the environment and programmes adult stress responsivity (Francis & Meaney, 1999). The standard laboratory environment, while not particularly enriching, is not very stressful. Therefore, adulthood anxiety is more likely to be driven by genetic or sex differences rather than breeding scheme in these particular conditions. In experiments where the environment elicits more stress (such as genetic manipulations resulting in reduced maternal care) then environmental manipulations, like breeding schemes, may have a greater effect on adulthood emotional behaviour. Wright & Brown (2000) found that sires increased pup care and survival but only in adverse environmental conditions. In humans, maternal depression during infancy reduces the level of maternal care received by the infant and results in poor cognitive development and emotion dysregulation (Mezulis, Hyde, & Clark, 2004). Paternal care can provide a buffer for these detrimental effects (Tannenbaum & Forehand, 1994). Father involvement has been found to

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

moderate effects of maternal depression during a child's infancy on later anxiety (Mezulis et al., 2004).

Presence of sires could augment or attenuate the effects of other environmental conditions during early development, causing long lasting effects. Future research should examine the effects of breeding scheme in different animal models, such as high anxiety (HAB) or low maternal care. Infant and adult measures of anxiety often only correlate in HAB models (Wohr & Schwarting, 2008). Therefore, presence of sires in infancy may produce longer lasting effects on adult emotional behaviour in HAB offspring. Previous work in our lab has found that SERT-/- KO animals have reduced maternal care. If the current study was reproduced in these animals, then the presence of sires during development may also have a greater impact on adult anxiety. Sires may contribute enough LG to make up for maternal care deficits in SERT KO, resulting in similar adult offspring emotional behaviour as seen in offspring from wild-type dams.

Conclusion. The current study shows that presence of the sire during early development of rats has an influence on infant and adolescence but not adulthood emotional behaviour. It suggests that the positive impact of sires on young pups is not strong enough to have a lasting effect into adulthood in wild-type rats bred under stress-free conditions. However, these effects do warrant further investigation under different genetic and environmental circumstances. Finally, this study produced some novel findings using the mSAT, suggesting that it is subject to sex differences as seen in many other anxiety behavioural paradigms.

It is important to understand how the presence of sires in early development affects infant, adolescent, and adulthood anxiety. The findings of the current study and proposed future research may suggest that continuous and intermittent breeding schemes can be used as

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

a tool to control for different levels of LG received by pups in different experimental manipulations.

References

- Allen, K., Dwinell, M., Zappa, A., Temple, A., & Thulin, J. (2013). Comparison of two rat breeding schemes using conventional caging. *Journal of the American Association for Laboratory Animal Science*, 52(2), 142-145.
- Barker, T., George, R., Howarth, G., & Whittaker, A. (2017) Assessment of housing density, space allocation and social hierarchy of laboratory rats on behavioural measures of welfare. *PloS ONE*. DOI: 10.1371/journal.pone.0185135
- Bateson, P. (1994) The dynamics of parent-offspring relationships in mammals. *Trends in Ecology & Evolution*, 9(10), 399-403. DOI: 10.1016/0169-5347(94)90066-3.
- Bosch, O. & Neumann, I. (2008). Brain vasopressin is an important regulator of maternal behaviour independent of dam's trait anxiety. *PNAS*, 105(4), 17139-17144. DOI: 10.1073/pnas.0807412105.
- Branchi, I., Santucci, D., & Alleva, E. (2001). Ultrasonic vocalizations emitted by infant rodents: A tool for assessment of neurobehavioural development. *Behavioural Brain Research*, 125(1-2), 49-56.
- Broad, K. D., Curley, J. P., & Keverne, E. B. (2006). Mother-Infant Bonding and the Evolution of Mammalian Social Relationships. *Philosophical Transactions: Biological Sciences*, 361(1476), 2199-2214.
- Brown, R. (1986). Social and hormonal factors influencing infanticide and its suppression in adult male Long Evans rats (*Rattus norvegicus*). *Journal of Comparative Psychology*, 100(2), 155-161. DOI: 10.1037/0735-7036.100.2.155
- Brumer, D., & Buinot, M. (1999). Anxiety, motor activation and maternal infant interactions in 5HT_{1B} knockout mice. *Behavioural Neuroscience*, 113(3), 587-601.

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

- Brummelte, S., & Galea, L. (2010) Chronic corticosterone during pregnancy and postpartum affects maternal care, cell proliferations and depressive-like behaviour in the dam. *Hormones & Behaviour*, 58(5), 769-779. DOI: 10.1016/j.yhbeh.2010.07.012.
- Brunelli, S. (2005) Selective breeding for infant phenotype: Rat pup ultrasonic vocalisation (USV). *Behavioural Genetics*, 35, 53-65.
- Brunelli, S., Masmela, J., Shair, H., & Hofer, M. (1998) Effects of biparental rearing on ultrasonic vocalisations (USV) responses of rat pups. *Journal of Comparative Psychology*, 112(4), 331-343.
- Caldji, C., Tannenbaum, B., Sharma, S., Francis, D., Plotsky, P., & Meaney, M. (1998) Maternal care during infancy regulate the development of neural systems mediating the expression of fearfulness in the rat. *Neurobiology*, 95, 5335-5340.
- Carobrez, A., & Bertoglio, L. (2005). Ethological and temporal analyses of anxiety-like behaviour: The elevated plus-maze model 20 years on. *Neuroscience and Behavioural Reviews*, 29, 1193-1205. DOI: 10.1016/j.neubiorev.2005.04.017
- Champagne, F., Francis, D., Mor, A., & Meaney, M. (2003) Variations in maternal care in the rat as a mediating influence for the effects of environment on development. *Physiology & Behaviour*, 79(3), 359-371. DOI: 10.1016/S0031-9384(03)00149-5.
- Conner, J., & Davis, H. (1980) Postpartum estrus in Norway rats. I. Behaviour. *Biology of Reproduction*, 23(5), 994-999. DOI: 10.1095/biolreprod23.5.994.
- Darlene, F., Champagne, F., Liu, D., & Meaney, M. (1999). Maternal care, gene expression, and the development of individual differences in stress reactivity. *Annals of the New York Academy of Sciences*, 896(1), 66-84.
- Deacon, R. (2013) The successive alleys test of anxiety in mice and rats. *Journal of Visualised Experiments*, 76, e2705.

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

- Denenberg, V., & Whimby, A. (1963). Behaviour of adult rats is modified by the experiences their mothers had as infants. *Science*, *142*, 1192-1193. DOI: 10.1126/science.142.3596.1192
- Dewsbury, D. (1985) Parental behaviour in rodents. *American Zoologist*, *25*, 841-852.
- Doremus, T., Brunell, S., Varlinskaya, E., & Spear, L. (2003). Anxiogenic effects during withdrawal from acute ethanol in adolescent and adult rats. *Pharmacology, Biochemistry, and Behaviour*, *75*, 411-418
- Doremus-Fitzwater, T., Varlinskaya, E., & Spear, L. (2009). Effects of pretest manipulation on elevated plus maze behaviour in adolescent and adult male and female Sprague-Dawley rats. *Pharmacology, Biochemistry, and Behaviour*, *92*, 413-423.
- Ehret, G. (2005) Infant rodent ultrasounds - A guide to the understanding of sound communication. *Behavioural Genetics*, *35*(1), 19-29.
- Ehret, G., & Haack, B. (1984). Motivation and arousal influence sound-induced maternal pup-retrieving behaviour in lactating house mice. *Ethology*, *65*(1), 25-39.
- Fernandez, C., Gonzales, M., Wilson, C., & File, S. (1999) Factor analysis shows that female rat behaviour is characterised primarily by activity, males by sex and anxiety. *Pharmacology, Biochemistry and Behaviour*, *64*(4), 731-736. DOI: 10.1016/S0091-3057(99)00139-2.
- Francis, D., Diorio, J., Liu, D., & Meaney, M., (1999) Nongenomic transmission across generations of maternal behaviour and stress responsiveness in the rat. *Science*, *286*(5442), 1155-1158. DOI: 10.1126/Science.286.5442.1155.
- Francis, D., Diorio, J., Plotsky, P., & Meaney, M. (2002). Environmental enrichment reverses the effects of maternal separation on stress reactivity. *Journal of Neuroscience*, *22*(18), 7840-7843. DOI: 10.1523/JNEUROSCI.22-18-07840.2002

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

Francis, D., & Meaney, M. (1999) Maternal care and the development of stress responses.

Current Opinion in Neurobiology, 9(1), 128-134. DOI: 10.1016/S0950-4388/99S80016-6.

Fujimo, H., Liu, H., Lopatina, O., Brown, D., & Higashida, H. (2013) Scopolamine modulates paternal parental retrieval behaviour in mice induced by the maternal mate.

Neuroscience Letters, 546, 63-66. DOI: 10.1016/j.neulet.2013.04.059.

Graham, L., Yoon, T., Yee, H., & Kim, J. (2009) Strain and sex differences in fear conditioning: 22kHz ultrasonic vocalisations and freezing in rats. *Psychology & Neuroscience*, 2, 219-225. DOI: 10.3922/j.psns.2009.2.015

Heim, C., & Nemeroff, C., (2001) The role of childhood trauma in the neurobiology of mood and anxiety disorders: Preclinical and clinical studies. *Biological Psychiatry*, 49(12), 1023-1039. DOI: 10.1016/S0006-3223(01)01157-X.

Hofer, M. (1996). Multiple regulators of ultrasonic vocalisations in the infant rat.

Psychoneuroendocrinology, 21(2), 203-217. DOI: 10.1016/0306-4530(95)00042-9.

Hofer, M., Masmela, J., Brunelli, S., & Shair, H. (1997). The ontogeny of maternal potentiation of the infant rats isolation call. *Developmental Psychobiology*, 33(3), 189-201. DOI: 10.1002/(SICI)1098-2302(199811)33:3

Insel, T., & Hill, J. (1987). Infant separation distress in genetically fearful rats. *Biological Psychiatry*, 22, 786-789.

Kuroda, K., Tavhikawa, K., Yoshida, S., Tsuneoka, Y., & Numan, M. (2011).

Neuromolecular basis of parental behaviour in laboratory mice and rats: With special emphasis on technical issues of using mouse genetics. *Progress in Neuropsychopharmacology & Biological Psychiatry*, 35(5), 1205-1231. DOI: 10.1016/j.pnpbp.2011.02.008.

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

- Li, L., Kevernes, E., Aparicio, A., Ishino, F., Barton, C., & Surani, M. (1999) Regulation of maternal behaviour and offspring growth by paternally expressed Peg3. *Science*, 284(5412), 330-334. DOI: 10.1126/science.284.5412.330.
- Liu, D., Diorio, J., Day, J., Francis, D, Meaney, M. (2000). Maternal care, hippocampal synaptogenesis and cognitive development in rats. *Nature Neuroscience*, 3(8), 799-806.
- Liu, H., Diorio, J., Tannenbaum, B., Caldji, C., Francis, D., Freedman, A., (...) & Meaney, M. (1997) Maternal care, hippocampal glucocorticoid receptor gene expression and hypothalamic pituitary adrenal responses to stress. *Science*, 277, 1659-1662. DOI: 10.1126/science.277.5332.1659.
- Liu, H., Lopatina, O., Higashida, C., Fujimoto, H., Akther, S., Inzhutova, A., ... Higashida, H. (2013). Displays of paternal mouse pup retrieving following communicative interaction with maternal mates. *Nature Communications*, 4(1346). DOI: 10.1038/ncomms2336.
- Lynn, D., & Brown, G. (2010). The ontogeny of anxiety-like behaviour in rats from adolescence to adulthood. *Developmental Psychobiology*, 52(8), 731-739. DOI: 10.1002/dev.20468.
- Mezulis, A., Hyde, J. & Clark, R. (2004) Father involvement moderates effect of maternal depression during a child's infancy on child behaviour problems in kindergarten. *Journal of Family Psychology*, 18(4), 575-588. DOI: 10.1037/0893-3200.18.4.575.
- Pederson, C., Vadlamundi, S., Boclia, M., & Amico, J. (2006). Maternal behaviour deficits in nulliparous oxytocin knockout mice. *Genes, Brain and Behaviour*, 5, 274-281. DOI: 10.1111/j.1601-183X.2005.00162.x

- Perry, R., & Sullivan, R. (2014) Neurobiology of attachment to an abusive caregiver: Short term benefits and long term costs. *Developmental Psychobiology*, 56(8), 1626-1634. DOI: 10.1002/dev.21219.
- Priebe, K., Brake, W., Romeo, R., Sisti, H., Mueller, A., McEwen B., & Francis, D. (2005) Maternal influences on adult stress and anxiety-like behaviour in C5BL/6J and BALB/cJ mice: A cross fostering study. *Developmental Psychobiology*, 47(4), 398-407. DOI: 10.1002/dev.20098.
- Priestel, R., & Young, S. (1978) An observational study of caretaking behaviour in male and female mice housed together. *Developmental Psychobiology*, 11, 23-30. DOI: 10.1002/dev.420110107
- Prusator, D., & Meerveld, G. (2015) Gender specific effects of neonatal limited nesting on viscerosomatic sensitivity and anxiety-like behaviour in adult rats. *Neurogastroenterology & Motility*, 27(1), 78-81. DOI: 10.1111/nmo.12472.
- Pryce, C., & Feldon, J., (2003) Long-term neurobehavioural impact of the postnatal environment in rats: Manipulations, effects and mediating mechanisms. *Neuroscience & Neurobehavioural Reviews*, 27(1-2), 57-71. DOI: 10.1016/S0149-7634(03)00009-5.
- Reistart, I., Karolczak, M., Beyer, C., Just, W., Maxso, S., & Ehret, G. (2002). Sry does not fully sex-reverse female into male behaviour towards pups. *Behaviour Genetics*, 32(2), 103-111.
- Rodel, H., Prager, G., Stefanski, V., Holst, D., & Hudson, R. (2008) Separating maternal and litter size effects on early postnatal growth in two species of altricial small mammals. *Physiology & Behaviour*, 93(4-5), 826-834.

- Schwarting, R. & Wohr, M. (2011) On the relationships between ultrasonic calling and anxiety related behaviour in rats. *Brazilian Journal of Medical and Biological Research*, 45(4), 337-348. DOI:10.1590/S0100-879X2012007500038.
- Shair, H., Muller, J., & Moore, H. (2008a). Dopamine's role in social modulation of infant isolation-induced vocalisation: I Reunion responses to the dam, but not littermates, are dopamine dependent. *Developmental Psychology*, 51(2), 131-46, DOI: 10.1002/dev.20353.
- Shair, H., Muller, J., & Moore, H. (2008b). Dopamine's role in social modulation of infant isolation-induced vocalisation: II Maternally modulated infant separation responses are regulated by D1 and D2 family dopamine receptors. *Developmental Psychobiology*, 51(2), 158-72, DOI:10.1002/dev.20355
- Tannenbaum, L., & Forehand, R. (1994). Maternal depressive mood: The role of the father in preventing adolescent problem behaviours. *Behaviour Research and Therapy*, 32, 321-325.
- Walker, F., Naicker, S., Hinwood, M., Dunn, N., & Days, T. (2009). Strain differences in coping behaviour, novelty seeking behaviour, and susceptibility to socially conditioned fear: A comparison between Wistar and Sprague Dawley rats. *Stress*, 12, 507-516. DOI: 10.3109/10253890802673134
- Whishaw, I. Q., & Kolb, B. (2014). *The Behavior of the Laboratory Rat : A Handbook with Tests*. New York, UNITED STATES: Oxford University Press.
- Wohr, M., Houx, B., Schwarting, R., & Spruijt, B. (2008) Effects of experience and context on 50kHz vocalizations in rats. *Physiology and Behaviour*, 93(4-5), 766-776. DOI: 10.1016/j.physbeh.2007.11.031
- Wohr, M., & Schwarting, R. (2008) Maternal care, isolation induced infant ultrasonic calling and their relationship to adult anxiety, *Behavioural Neuroscience*, 122(2), 310-330.

DO MALE RATS SHARE IN THE PARENTAL CARE OF PUPS?

- Wright, S., & Brown, R. (2000). Maternal behaviour, paternal behaviour, and pup survival on CD-1 albino mice (*Mus musculus*) in 3 different housing conditions. *Journal of Comparative Psychology*, *114*(2), 183-192. DOI: 10.1037/0735-7036.114.2.183.
- Yee, N., Schwarting, R., Fuchs, E., & Wohr, M. (2012) Juvenile stress potentiates aversive 22kHz ultrasonic vocalisations and freezing during auditory fear conditions in adult male rats. *Stress*, *15*(5). DOI: 10.3109/10253890.2011.646348.